

TASK-1

```
1  #include<iostream>
2  using namespace std;
3  class transaction{
4      protected:
5          string  accNumber;
6          double Balance;
7      public:
8          transaction(string num,double a){
9              accNumber = num;
10             Balance = a;
11         }
12         virtual void processTrasaction(double a)=0;
13     };
14
15
16
17 class deposit :public transaction{
18     public:
19     deposit(string num,double a ):transaction(num,a){}
20
21     void processTrasaction(double a){
22         if(a >= 0){
23             Balance += a;
24             cout<<"\n----Deposit Transaction----\n";
25             cout<<"Account Number: "<<accNumber<<endl;
26             cout<<"Amount: "<<a<<endl;
27             cout<<"Status: Amount Deposited Successfully"<<endl;
28             cout<<"New Balance: "<<Balance<<endl;
29         }else{
30             cout<<"Amount Must be Positive!\n";
31         }
32         cout<<"\n-----\n";
33     }
34 };
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56 class Withdraw: public transaction{
57     public:
58     Withdraw(string num,double a ):transaction(num,a){}
59
60     void processTrasaction(double a){
61         if(Balance >= a){
62             Balance -= a;
63             cout<<"\n----Withdraw Transaction----\n";
64             cout<<"Account Number: "<<accNumber<<endl;
65             cout<<"Amount: "<<a<<endl;
66             cout<<"Status: Amount Withdrawn Successfully"<<endl;
67             cout<<"New Balance: "<<Balance<<endl;
68         }else{
69             cout<<"Balance Insufficient!\n";
70         }
71         cout<<"\n-----\n";
72     }
73 };
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65 class transfer: public transaction{
66     public:
67     transfer(string num,double a ):transaction(num,a){}
68
69     void processTrasaction(double a){
70         string acc;
71         cout<<"Enter Account Number To Transfer: ";
72         cin>>acc;
73         if(Balance >= a){
74
75             Balance -= a;
76             cout<<"\n---Transfer Transaction---\n";
77             cout<<"Reciever Account Number: "<<accNumber<<endl;
78             cout<<"Sender Account Number: "<<acc<<endl;
79             cout<<"Amount: "<<a<<endl;
80             cout<<"Status: Amount Transfered Successfully"<<endl;
81             cout<<"New Balance: "<<Balance<<endl;
82         }else{
83             cout<<"Balance Insufficient!\n";
84         }
85         cout<<"\n-----\n";
86     }
87 };
```

```
79 int main(){
80     transaction *t;
81     deposit d1("PXBX00981002",12000);
82     t= &d1;
83     t->processTrasaction(100);
84
85     Withdraw w1("PXBTY3340",13000);
86     t =&w1;
87     t->processTrasaction(300);
88
89     transfer tf("PXYB6679",14000);
90     t =&tf;
91     t->processTrasaction(14100);
92 }
```

```
----Deposit Transaction----
Account Number: PXXB00981002
Amount: 100
Status: Amount Deposited Successfully
New Balance: 12100

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----Withdraw Transaction----
Account Number: PXXBTY3340
Amount: 300
Status: Amount Withdrawn Successfully
New Balance: 12700

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Enter Account Number To Transfer: 2500
Balance Insufficient!

-----
```

TASK-2

```
1  #include <iostream>
2  using namespace std;
3
4  class Notification
5  {
6  protected:
7      string recipient;
8
9  public:
10     Notification(string recipient)
11     {
12         this->recipient = recipient;
13     }
14     virtual void sendNotification() = 0;
15 };
16 class EmailNotification : public Notification
17 {
18 public:
19     EmailNotification(string recipient) : Notification(recipient) {}
20     void sendNotification()
21     {
22         cout << "\n----Email Notification----\n";
23         cout << "To: " << recipient << endl;
24         cout << "Your OTP is 123456. Please do not share it with anyone." << endl;
25         cout << "Status: Sent successfully!" << endl;
26         cout << "-----\n";
27     }
28 };
29
```

```
29 class SMSNotification : public Notification
30 {
31 public:
32     SMSNotification(string recieipient) : Notification(recieipient) {}
33     void sendNotification()
34     {
35         cout << "\n---SMS Notification---\n";
36         cout << "To: " << recieipient << endl;
37         cout << "Your OTP is 123456. Please do not share it with anyone." << endl;
38         cout << "Status: Sent successfully!" << endl;
39         cout << "-----\n";
40     }
41 };
42 class PushNotification : public Notification
43 {
44 public:
45     PushNotification(string recieipient) : Notification(recieipient) {}
46     void sendNotification()
47     {
48         cout << "\n---Push Notification---\n";
49         cout << "To: " << recieipient << endl;
50         cout << "Your OTP is 123456. Please do not share it with anyone." << endl;
51         cout << "Status: Sent successfully!" << endl;
52         cout << "-----\n";
53     }
54 };
```

```
55 int main()
56 {
57     Notification *n1;
58     EmailNotification email("qasim@gmail.com");
59     SMSNotification sms("03343329938");
60     PushNotification push("Mahboor Ali");
61
62     n1 = &email;
63     n1->sendNotification();
64
65     n1 = &sms;
66     n1->sendNotification();
67
68     n1 = &push;
69     n1->sendNotification();
70 }
```

----Email Notifcation----

To: qasim@gmail.com

Your OTP is 123456. Please do not share it with anyone.

Status: Sent successfully!

----SMS Notifcation----

To: 03343329938

Your OTP is 123456. Please do not share it with anyone.

Status: Sent successfully!

----Push Notifcation----

To: Mahboor Ali

Your OTP is 123456. Please do not share it with anyone.

Status: Sent successfully!
